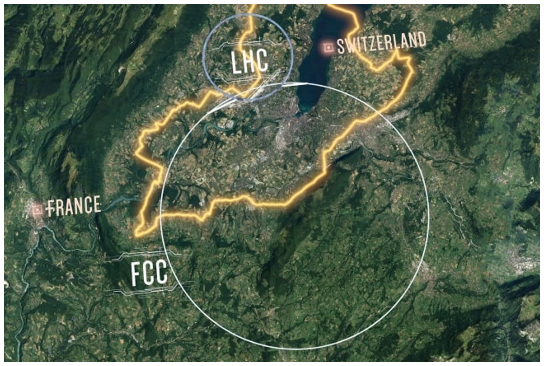
Parametrizing Curves in 2D and 3D

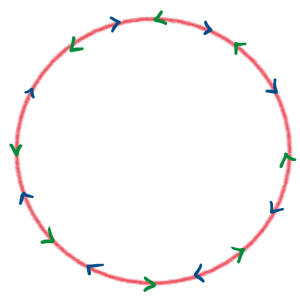
Tuesday, May 6, 2025 6:45 AM



1 Parameterizing a Path

1.1 What do you see?

Large Hadron Collider - modern Engineering marvel!

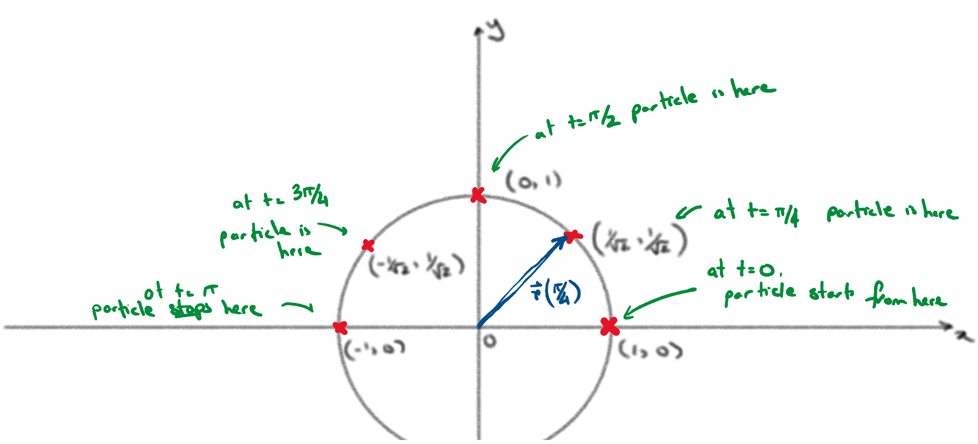


Particles whizzing around in circles. need to track/control their movements

1.2 How do you describe the path taken by a particle?

- use position vector and parametrization
- Track the location of the particle at all times t and use a function $\vec{r}(t)$ to describe it.

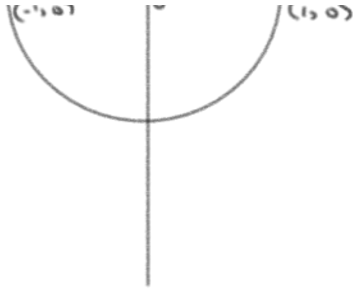
1.3 Example:



- particle's path traces out portion of the unit circle
- Equation of unit circle $x^2+y^2=1$
- $\vec{r}(t)$ is a vector having two functions.

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

such that $x(t)$ gives



$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$

Such that $x(t)$ gives the x-position at time t (similarly $y(t)$ is y-position at time t)

-need to make sure:

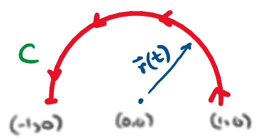
$[x(t)]^2 + [y(t)]^2 = 1$

Answer: $x(t) = \cos(t)$

$y(t) = \sin(t)$

So that $\vec{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$, i.e. $\vec{r}(t) = \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$ for $0 \le t \le \pi$.

Overall:



1.4 Parametric representations induce orientation on C:

as we increase t, we travel along C in a certain direction.

1.5 Quick Questions:

→ How would the parametrization of a circle change if:

- radius was changed from 1 to 2, etc.
- the center was shifted from (0, 0) to (3, 4)?

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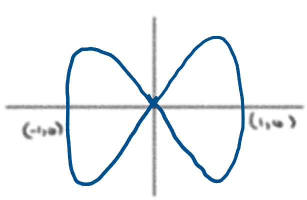
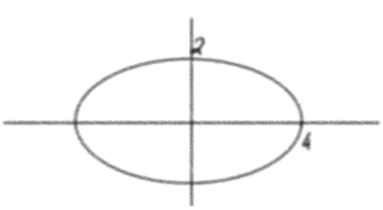
1.6 Quick Questions.

Match the curve with its parametrization.

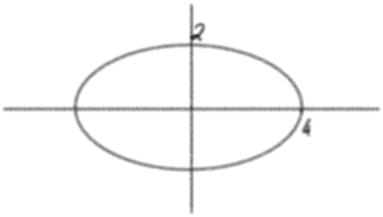


(A) $(4\cos t, 2\sin t)$, $0 \le t \le 2\pi$



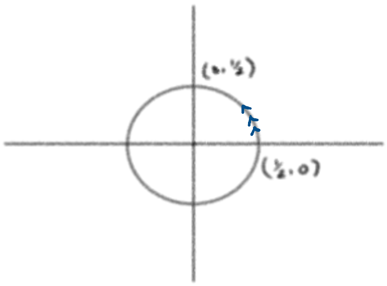
(B) $\left(\frac{1}{2}\cos(3t), \frac{1}{2}\sin(3t)\right)$

$0 \le t \le 2\pi$



(B) $\left(\frac{1}{2}\cos(3t),\ \frac{1}{2}\sin(3t)\right)$

$0 \le t \le 2\pi$



(C) $\left(\cos(2t),\ \sin(4t)\right)$

$0 \le t \le 2\pi$

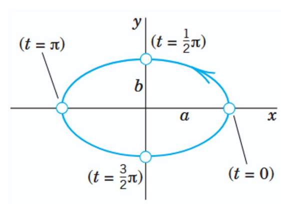
1.7 Some common parameterizations.

The vector function

(3) $$\mathbf{r}(t) = [a\cos t,\quad b\sin t,\quad 0] = a\cos t\,\mathbf{i} + b\sin t\,\mathbf{j} \qquad \text{(Fig. 202)}$$

represents an ellipse in the *xy*-plane with center at the origin and principal axes in the direction of the *x*- and *y*-axes. In fact, since $\cos^2 t + \sin^2 t = 1$, we obtain from (3)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad z = 0.$$

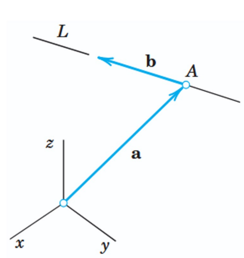


A straight line L through a point A with position vector $\mathbf{a}$ in the direction of a constant vector $\mathbf{b}$ (see Fig. 203) can be represented parametrically in the form

(4) $$\mathbf{r}(t) = \mathbf{a} + t\mathbf{b} = [a_1 + tb_1,\quad a_2 + tb_2,\quad a_3 + tb_3].$$

For instance, the straight line in the *xy*-plane through A: (3, 2) having slope 1 is (sketch it)

$$\mathbf{r}(t) = [3,\quad 2,\quad 0] + t[1,\quad 1,\quad 0] = [3 + t,\quad 2 + t,\quad 0].$$



How did they get this ?

1.8 To find equation of a straight line:

- find two points $A(a_1, a_2, a_3)$ and $B(p_1, p_2, p_3)$

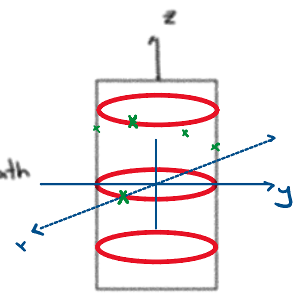
1.8 To find equation of a straight line:

- find two points $A(a_1, a_2, a_3)$ and $B(p_1, p_2, p_3)$
- then $\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$, $\vec{b} = \vec{AB} = \begin{bmatrix} p_1 - a_1 \\ p_2 - a_2 \\ p_3 - a_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$
- parametrization from A to B:

$$\vec{r}(t) = \vec{a} + t\vec{b}.$$

1.9 What curve C do we get from the parametrization:

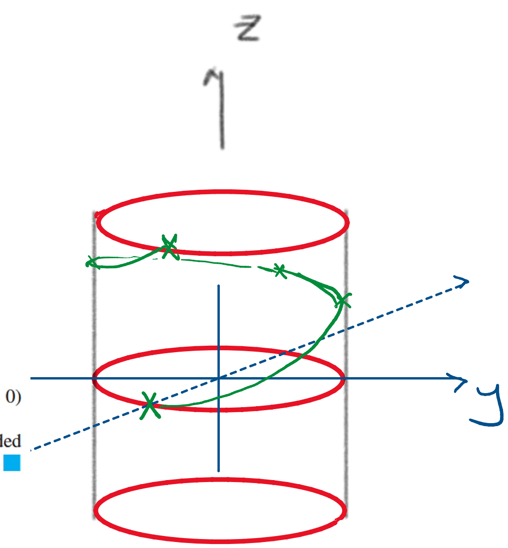
$$\vec{r}(t) = \begin{bmatrix} a\cos t \\ a\sin t \\ ct \end{bmatrix}, \quad 0 \le t \le 2\pi \ ?$$



Soln: $\left.\begin{array}{l} x(t) = a\cos t \\ y(t) = a\sin t \end{array}\right\} \Rightarrow x^2 + y^2 = a^2 \leftarrow$ traces a circular path

but $z(t) = ct \Rightarrow$ height increases with t

t	x	y	z
0	a	0	0
$\pi/4$	$a/\sqrt{2}$	$a/\sqrt{2}$	$c\pi/4$
$\pi/2$	0	a	$c\pi/2$
$3\pi/4$	$-a/\sqrt{2}$	$a/\sqrt{2}$	$3c\pi/4$
π	$-a$	0	$c\pi$
$3\pi/2$	0	$-a$	$3c\pi/2$
2π	a	0	$2c\pi$



The twisted curve C represented by the vector function

$$\mathbf{r}(t) = [a\cos t, \quad a\sin t, \quad ct] = a\cos t\,\mathbf{i} + a\sin t\,\mathbf{j} + ct\,\mathbf{k} \qquad (c \neq 0) \tag{5}$$

is called a *circular helix*. It lies on the cylinder $x^2 + y^2 = a^2$. If $c > 0$, the helix is shaped like a right-handed screw (Fig. 204). If $c < 0$, it looks like a left-handed screw (Fig. 205). If $c = 0$, then (5) is a circle. ■

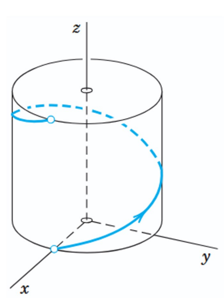


Fig. 204. Right-handed circular helix

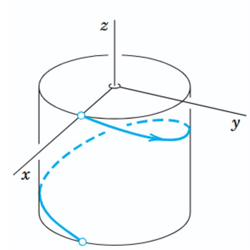
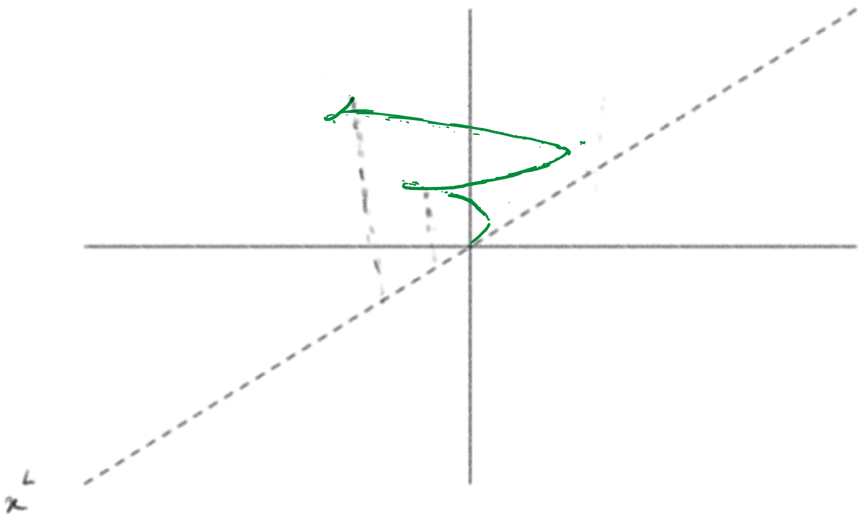
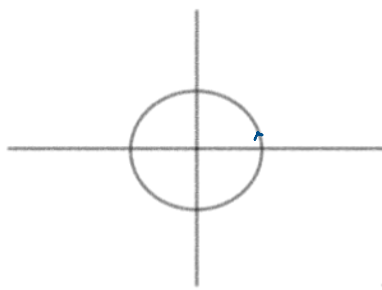


Fig. 205. Left-handed circular helix

1.10 Exercise: Try to parameterize the following.



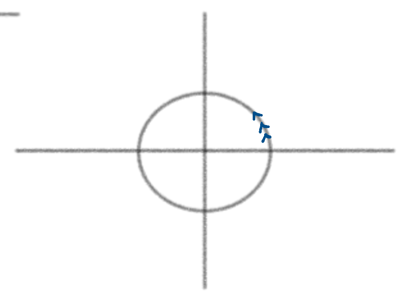
2. Velocity, Speed, and Acceleration



2.1 What is the difference between

$$\vec{r}(t) = \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$$

and $\vec{r}(t) = \begin{bmatrix} \cos(3t) \\ \sin(3t) \end{bmatrix}$ for $0 \le t \le 2\pi$?



both are circles, but one winds around 3-times while the other only once. The difference is in velocity and acceleration.

2.2

$$\mathbf{v}(t) = \mathbf{r}'(t), \qquad \mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t).$$

2.3 Example:

for $\vec{r} = \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$, $\vec{r}'(t) = \begin{bmatrix} \frac{d}{dt}\cos t \\ \frac{d}{dt}\sin t \end{bmatrix} = \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix}$

*Remember trig derivatives

For $\vec{r} = \begin{bmatrix} \cos 3t \\ \sin 3t \end{bmatrix}$, $\vec{r}'(t) = \begin{bmatrix} -3\sin 3t \\ 3\cos 3t \end{bmatrix} = 3\begin{bmatrix} -\sin(3t) \\ \cos(3t \end{bmatrix}$

2.4 Exercise: Find the acceleration for the above curves.

2.5 Suppose at some time t_0, a path $r(t)$ along C is at point P given by $\vec{r}_0$, with velocity $(\vec{r}')_0$. Note: $\vec{r}_0$ is just $\vec{r}(t_0)$
and $(\vec{r}')_0$ is just $\vec{r}'(t_0)$

numbers/exact vectors

Then the tangent line to C at P is given by the equation:

$$\vec{q}(w) = \vec{r}_0 + w(\vec{r}\,')_0$$

INSERT Rant about textbook notation.

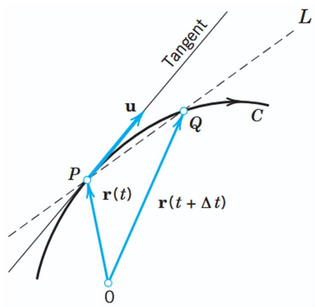


Fig. 207. Tangent to a curve

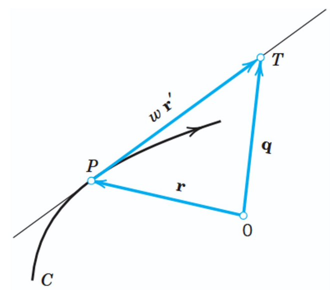


Fig. 208. Formula (9) for the tangent to a curve

2.6 **Tangent to an Ellipse**

Find the tangent to the ellipse $\frac{1}{4}x^2 + y^2 = 1$ at P: $(\sqrt{2}, 1/\sqrt{2})$.

Solution. Equation (3) with semi-axes $a = 2$ and $b = 1$ gives $\mathbf{r}(t) = [2\cos t, \sin t]$. The derivative is $\mathbf{r}'(t) = [-2\sin t, \cos t]$. Now P corresponds to $t = \pi/4$ because

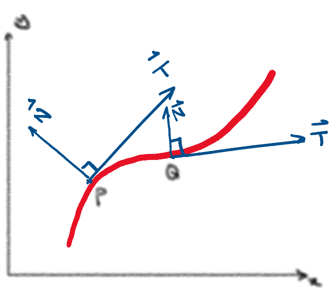
$$\mathbf{r}(\pi/4) = [2\cos(\pi/4), \sin(\pi/4)] = [\sqrt{2}, \quad 1/\sqrt{2}].$$

Hence $\mathbf{r}'(\pi/4) = [-\sqrt{2}, \quad 1/\sqrt{2}]$. From (9) we thus get the *answer*

$$\mathbf{q}(w) = [\sqrt{2}, \quad 1/\sqrt{2}] + w[-\sqrt{2}, \quad 1/\sqrt{2}] = [\sqrt{2}(1 - w), \quad (1/\sqrt{2})(1 + w)].$$

To check the result, sketch or graph the ellipse and the tangent. ■

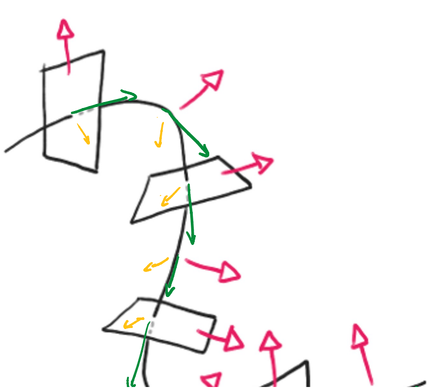
2.7 The Frenét frame



In 2D. can form a "frame" by using tangent and normal vector at each point. Think of these as "alternate axes".

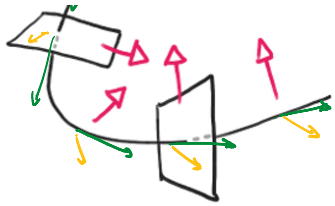
any vector can be written as a combination of $\vec{T}$ and $\vec{N}$ at P.

2.8



In 3D there is an entire plane that is normal to the tangent vector.

Engineering is often interested in the tangential and normal components of a vector field along a curve.



the tangential and normal components of a vector field along a curve.
(eg. gravitation field along path of a rocket, etc)

2.9 The velocity vector $\vec{v}(t_0) = \vec{r}\,'(t_0)$ (we'll just say $\vec{v}$) is a <u>direction</u> for the <u>tangent vector</u>. For another vector $\vec{E}$, we get

$$\vec{E}_{tan} = \frac{\vec{E}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\,\vec{v} \qquad \text{and} \qquad \vec{E}_{norm} = \vec{E} - \vec{E}_{tan}$$

tangential Component

Normal Component

NOTE: Review dot product (pg 361) and projection (pg 365)

Inner Product (Dot Product) of Vectors

The **inner product** or **dot product** $\mathbf{a}\bullet\mathbf{b}$ (read "**a** dot **b**") of two vectors **a** and **b** is the product of their lengths times the cosine of their angle (see Fig. 178),

$$\textbf{(1)} \qquad \begin{aligned} \mathbf{a}\bullet\mathbf{b} &= |\mathbf{a}||\mathbf{b}|\cos\gamma && \text{if} \quad \mathbf{a}\neq\mathbf{0}, \mathbf{b}\neq\mathbf{0} \\ \mathbf{a}\bullet\mathbf{b} &= 0 && \text{if} \quad \mathbf{a}=\mathbf{0} \text{ or } \mathbf{b}=\mathbf{0}. \end{aligned}$$

The angle γ, $0 \leqq \gamma \leqq \pi$, between **a** and **b** is measured when the initial points of the vectors coincide, as in Fig. 178. In components, $\mathbf{a} = [a_1, a_2, a_3]$, $\mathbf{b} = [b_1, b_2, b_3]$, and

$$\textbf{(2)} \qquad \mathbf{a}\bullet\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

2.10 We are particularly interested in the acceleration (Thanks Newton!) of the curve C, ie, $\vec{a} = \vec{v}\,'(t_0) = \vec{r}\,''(t_0)$ and its <u>decomposition</u> into <u>tangential</u> and normal <u>components</u>.

$$\mathbf{a}_{\text{tan}} = \frac{\mathbf{a}\bullet\mathbf{v}}{\mathbf{v}\bullet\mathbf{v}}\,\mathbf{v}. \qquad \text{Also,} \qquad \mathbf{a}_{\text{norm}} = \mathbf{a} - \mathbf{a}_{\text{tan}}.$$

<u>Reading</u>: Centripetal Acceleration (Example 7, pg 387)

EXAMPLE 7 **Centripetal Acceleration. Centrifugal Force**

The vector function

$$\mathbf{r}(t) = [R \cos \omega t, \quad R \sin \omega t] = R \cos \omega t\, \mathbf{i} + R \sin \omega t\, \mathbf{j} \qquad \text{(Fig. 210)}$$

(with fixed $\mathbf{i}$ and $\mathbf{j}$) represents a circle C of radius R with center at the origin of the xy-plane and describes the motion of a small body B counterclockwise around the circle. Differentiation gives the velocity vector

$$\mathbf{v} = \mathbf{r}' = [-R\omega \sin \omega t, \quad R\omega \cos \omega t] = -R\omega \sin \omega t\, \mathbf{i} + R\omega \cos \omega t\, \mathbf{j} \qquad \text{(Fig. 210)}$$

$\mathbf{v}$ is tangent to C. Its magnitude, the speed, is

$$|\mathbf{v}| = |\mathbf{r}'| = \sqrt{\mathbf{r}' \bullet \mathbf{r}'} = R\omega.$$

Hence it is constant. The speed divided by the distance R from the center is called the **angular speed**. It equals ω, so that it is constant, too. Differentiating the velocity vector, we obtain the acceleration vector

$$\mathbf{a} = \mathbf{v}' = [-R\omega^2 \cos \omega t, \quad -R\omega^2 \sin \omega t] = -R\omega^2 \cos \omega t\, \mathbf{i} - R\omega^2 \sin \omega t\, \mathbf{j}. \tag{19}$$

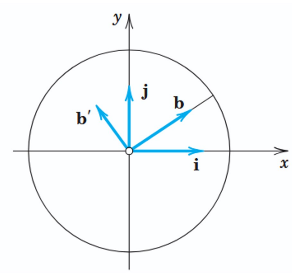


Fig. 210. Centripetal acceleration **a**

This shows that $\mathbf{a} = -\omega^2 \mathbf{r}$ (Fig. 210), so that there is an acceleration toward the center, called the **centripetal acceleration** of the motion. It occurs because the velocity vector is changing direction at a constant rate. Its magnitude is constant, $|\mathbf{a}| = \omega^2 |\mathbf{r}| = \omega^2 R$. Multiplying $\mathbf{a}$ by the mass m of B, we get the **centripetal force** $m\mathbf{a}$. The opposite vector $-m\mathbf{a}$ is called the **centrifugal force**. At each instant these two forces are in equilibrium.

We see that in this motion the acceleration vector is normal (perpendicular) to C; hence there is no tangential acceleration. ■